



اُنِيُوَرَسِيْتِي تِيكْنُولُوْجِي مَارَا
UNIVERSITI
TEKNOLOGI
MARA

Cawangan Terengganu
Kampus Kuala Terengganu

FACULTY OF COMPUTER

AND

MATHEMATICAL SCIENCES

ASSIGNMENT 1

SEMESTER	:	4
COURSE CODE	:	CSC584
COURSE NAME	:	ENTERPRISE PROGRAMMING
LECTURER	:	AHMAD AIMAN BIN SHALLAHUDDIN
SYSTEM NAME	:	CV WEB APPLICATION

STUDENT ID	NAME	GROUP
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DATE OF SUBMISSION	6/6/2026
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1.0 PROJECT OVERVIEW

This project is a Curriculum Vitae (CV) Generator Web Application developed as part of the CSC584 Enterprise Programming assignment. The application allows users to input their personal, educational, and professional information through an HTML form and generates a professionally formatted CV page. User also can see an example of generated CV so that they can know the outcome what they want. The application is built using the Model-View-Controller (MVC) architectural pattern with Java EE technologies:

Technology	Purpose
Java EE / Jakarta Servlets	Backend controller logic and HTTP request handling
JSP (JavaServer Pages)	Frontend view rendering with dynamic content
Tailwind CSS (CDN)	Responsive and professional UI styling
Apache Tomcat 10	Web server and servlet container
Maven	Project build and dependency management

2.0 MVC ARCHITECTURE EXPLANATION

The application strictly follows the Model-View-Controller (MVC) design pattern. This ensures a clean separation of concerns where no business logic exists inside the JSP views, and no HTML rendering occurs inside the Servlet controllers.

2.1 Model

2.1.1 cvData.java

The Model represents the data structure of the application. CvData.java is a Plain Old Java Object (POJO) that holds all CV-related data submitted by the user. It contains three inner classes to represent complex data structures:

- CvData - main class holding personal info (name, email, phone, location, LinkedIn, summary, job title)
- CvData.Education- inner class holding institution, degree, field of study, years, and notes
- CvData.Experience - inner class holding job title, company, dates, and responsibilities
- CvData.Project - inner class holding project title, technologies, and description

The Model has no knowledge of how data is displayed or received — it only stores and provides data through getters and setters.

2.2 View

The Views are JSP files responsible only for displaying data. They contain no business logic:

2.2.1 form.jsp

Displays the CV input form styled with Tailwind CSS. Sections include Personal Information, Education, Work Experience, Skills, and Projects. Supports dynamic addition of multiple entries via JavaScript.

2.2.2 cv.jsp

Displays the generated CV using data from the CvData object passed by the controller. Uses JSP scriptlets (`<%= %>`) to render dynamic content. Styled with a dark navy header and gold accents matching a professional CV design.

2.3 Controller

The Controllers are Jakarta Servlets that handle HTTP requests. They act as the bridge between the Model and the View:

2.3.1 CvFormServlet.java

- `CvFormServlet.doGet()` — receives a GET request and forwards to `form.jsp` to display the empty form
- `CvFormServlet.doPost()` — receives POST data from the form, reads all parameters, builds a `CvData` object, sets it as a request attribute, then forwards to `cv.jsp`

2.3.2 ExampleCvServlet.java

- `ExampleCvServlet.doGet()` — hardcodes a sample `CvData` object (Ahmad Zulkifli) and forwards to `cv.jsp`

3.0 FOLDER STRUCTURE

The project follows the standard Maven Web Application structure with proper MVC package organisation:

```
CurriculumVitae/
├── pom.xml
├── src/
│   ├── main/
│   │   ├── java/
│   │   │   ├── com/
│   │   │   │   ├── mycompany/
│   │   │   │   │   ├── curriculumvitae/
│   │   │   │   │   │   ├── model/
│   │   │   │   │   │   │   ├── CvData.java          ← (Model)
│   │   │   │   │   │   │   └── controller/
│   │   │   │   │   │   │       ├── CvFormServlet.java    ← (Controller)
│   │   │   │   │   │   │       └── ExampleCvServlet.java ← (Controller)
│   │   │   │   │   └── webapp/
│   │   │   │   │       ├── WEB-INF/
│   │   │   │   │       │   ├── web.xml
│   │   │   │   │       │   └── jsp/
│   │   │   │   │       │       ├── form.jsp             ← (View)
│   │   │   │   │       │       └── cv.jsp                ← (View)
│   │   │   │   │       └── index.jsp                    ← redirect page
```

4.0 CHALLENGES FACED

Challenge	Solution
Port 8080 already in use when starting Tomcat	Used Task Manager to identify and terminate the existing Java process occupying port 8080, then restarted the server successfully.
ClassNotFoundException: com.cvapp.controller.CvFormServlet	The web.xml servlet-class referenced the wrong package name (com.cvapp) while the actual Java files were under com.mycompany.curriculumvitae. Fixed by updating web.xml to match the correct package path.
Takes time to learn Java EE	Dedicated extra time for practice and followed online tutorials and documentation to improve understanding.
Project not appearing on GitHub at first	Learned how to properly initialize a repository, commit changes, and push code to GitHub using Github Desktop.

5.0 SCREENSHOTS

The following screenshots demonstrate the working application:

5.1 CV Input Form Page

CURRICULUM VITAE BUILDER

Craft Your Professional Story

Fill in your details below and generate a polished CV instantly.

[View Example CV](#)

PERSONAL INFORMATION

FULL NAME *
e.g. Ahmad Zulkifli

JOB TITLE *
e.g. Junior Java Developer

EMAIL ADDRESS *
you@email.com

PHONE NUMBER *
+60 12-345 6789

LOCATION
Kuala Lumpur, Malaysia

LINKEDIN / PORTFOLIO URL
linkedin.com/in/yourname

PROFILE SUMMARY
A brief overview of your professional profile, goals, and what makes you unique...

EDUCATION

INSTITUTION NAME *
Universiti Teknologi Malaysia

DEGREE / QUALIFICATION
Bachelor of Computer Science

FIELD OF STUDY
Software Engineering

START YEAR
2020

END YEAR
2024 (or Present)

NOTES / ACHIEVEMENTS
Dean's List, GPA 3.8, etc.

+ Add Another Education

WORK EXPERIENCE

JOB TITLE *
Software Engineer

COMPANY / ORGANISATION
TechCorp Sdn. Bhd.

START DATE
Jan 2022

END DATE
Dec 2023 (or Present)

KEY RESPONSIBILITIES & ACHIEVEMENTS
• Developed REST APIs using Spring Boot
• Reduced load time by 40% through caching

+ Add Another Experience

SKILLS

TECHNICAL SKILLS
e.g. Java, Spring Boot, MySQL (comma separated)

SOFT SKILLS
e.g. Leadership, Teamwork (comma separated)

PROJECTS

OPTIONAL

PROJECT TITLE
CV Builder Web App

TECHNOLOGIES USED
Java EE, JSP, Tailwind CSS

DESCRIPTION
Brief description of the project, your role, and outcome...

+ Add Another Project

[See an Example CV first](#)

Generate My CV →

5.2 Generated CV Page

[← Back to Form](#)[Print / Save PDF](#)

SENIOR JAVA DEVELOPER

Muhammad Amri Izwan

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0146489377

Puchong, Selangor

<https://github.com/amriizwan>

● Profile Summary

I am passionate about acquiring new skills and have developed strong critical thinking, problem-solving, and time management abilities through my coursework. I am seeking an opportunity where I can further develop my skills and knowledge while collaborating with a diverse group of people. I am eager to contribute my perspectives and ideas to help achieve the company's goals.

● Work Experience

○ IT Personnel

ZEN Apps Sdn Bhd

Apr 2025 — Oct 2025

IT Support / Helpdesk: Provide frontline technical assistance (Level 1/2) for hardware, software, and connectivity issues.

Network & Systems Administration: Set up, maintain, and secure servers, cloud infrastructure, and internal networks.

● Education

○ Universiti Teknologi Mara Kuala Terengganu

Bachelor of Computer Science — Computer Science

2025 — 2027

Dean's List

○ Universiti Teknologi Mara Segamat

Diploma in Computer Science — Computer Science

2022 — 2024

Anugerah Naib Canselor

● Projects

○ E-Learning System

Java EE, JSR, Tailwind CSS

An E-Learning System called MyStudyZone, is an online educational platform that enables students and lecturers to continue the learning process anywhere on any device when physical classes are not possible to establish. It functions as a centralized system, web-based client-server system where all learning data and activities are managed in a central server, allows users to access it through web browser and operates by interaction between clients (users) and server. A students can access course materials, assignments, while lecturers can organize and maintain student records.

TECHNICAL SKILLS

HTML/CSSMySQLJava

SOFT SKILLS

LeadershipTeamwork

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5.3 Example CV Page

[← Back to Form](#)[Print / Save PDF](#)

JUNIOR JAVA DEVELOPER

Ahmad Zulkifli bin Hassan

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+60 12-345 6789

Kuala Lumpur, Malaysia

linkedin.com/in/ahmadzulkifli

● Profile Summary

Motivated Computer Science graduate with hands-on experience in Java EE web development and a passion for building clean, scalable applications. Adept at applying MVC architecture, RESTful API design, and modern front-end frameworks to deliver impactful software solutions.

● Work Experience

○ Junior Java Developer

BulanBintang Sdn. Bhd.

Jan 2024 — Present

- Developed and maintained RESTful APIs using Java EE and Jakarta Servlets
- Implemented MVC architecture reducing codebase complexity by 30%
- Integrated Tailwind CSS front-end with back-end controller logic
- Participated in Agile sprint cycles and daily stand-ups

○ Software Engineering Intern

BulanMatahari Digital Services

Jun 2023 — Dec 2023

- Built a customer data management module using Spring Boot and MySQL
- Optimised SQL queries, improving report generation speed by 40%
- Wrote unit tests achieving 85% code coverage with JUnit

● Education

○ Universiti Teknologi Malaysia (UTM)

Bachelor of Computer Science — Software Engineering

2020 — 2024

Dean's List — Semester 3 & 5 | GPA 3.72 / 4.00

○ SMK Taman Desa

Sijil Pelajaran Malaysia (SPM)

2015 — 2019

IOA — Best Student Award

● Projects

○ CV Builder Web Application

Java EE, Jakarta Servlets, JSP, Tailwind CSS

A full-stack MVC web app enabling users to generate professional CVs via dynamic HTML forms.

○ E-Library Management System

Spring Boot, MySQL, React

Library system with book borrowing, user auth, and overdue notifications via JavaMail.

TECHNICAL SKILLS

Java EE, Jakarta Servlets, JSP, Spring Boot, MySQL, HTML/CSS, Tailwind CSS, Git

SOFT SKILLS

Problem Solving, Team Collaboration, Communication, Time Management, Adaptability

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6.0 GITHUB REPOSITORY

The complete source code for this project is available on GitHub:

Repository URL: <https://github.com/amriizwan/CurriculumVitae>

The repository contains:

- Complete Maven project source code
- All Java files (Model, Controllers)
- All JSP view files (form.jsp, cv.jsp)
- Configuration files (web.xml, pom.xml)
- README.md with setup and run instructions

7.0 CONCLUSION

This assignment successfully demonstrates the implementation of a Java EE web application following the MVC architectural pattern. The application achieves a clean separation of concerns with no logic mixing between layers — the Servlet controllers handle all data processing, the CvData POJO stores all data, and the JSP views are responsible only for rendering.

The use of Tailwind CSS via CDN enabled a professional and responsive UI design without requiring a Node.js build pipeline, making it suitable for a pure Java EE environment. The application supports multiple education entries, multiple work experience entries, and multiple projects, demonstrating dynamic form handling with JavaScript.

The example CV feature (ExampleCvServlet) demonstrates that the same cv.jsp template works correctly with both user-submitted data and hardcoded data, confirming the reusability and flexibility of the MVC design.